

Solving and Graphing One-Step Inequalities

Answer Key

Grade 7

Quick Answers

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|---|--|
| 1. (a) the solution = $(-\infty, 7)$, — | 7. (a) the solution = $(-\infty, 5]$, — |
| 2. (a) the solution = $[7, \infty)$, — | 8. (a) the solution = $(-\infty, -3)$, — |
| 3. (a) the solution = $(-\infty, 5]$, — | 9. (a) the solution = $(-8, \infty)$, — |
| 4. (a) the solution = $(-7, \infty)$, — | 10. (a) the solution = $(-\infty, 5]$, — |
| 5. (a) the solution = $(6, \infty)$, — | 11. (a) the solution = $(-7, \infty)$, — |
| 6. (a) the solution = $(-\infty, -5]$, — | 12. (a) the solution = $(-\infty, -6]$, — |
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What is verified

0 of 12 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 7

7.EE.B.4 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12

Difficulty 1–5 of 5 across 25 tagged checks.

1. Solve and graph $x + 5 < 12$

Solution: (a) Undo the +5 by subtracting 5 from both sides. Adding or subtracting never changes the direction of the sign: $x < 12 - 5$, so $x < 7$

(b) **Instructor-judged graph.** Open circle at 7, shaded to the left. Check: $6 + 5 = 11 < 12$ works, $7 + 5 = 12$ does not.

2. Solve and graph $y - 3 \geq 4$

Solution: (a) Add 3 to both sides: $y \geq 4 + 3$, so $y \geq 7$

(b) **Instructor-judged graph.** Filled circle at 7, shaded to the right. The circle is filled because $\geq$ allows equality, and $7 - 3 = 4$ exactly.

3. Solve and graph $4m \leq 20$

Solution: (a) Divide both sides by 4. Four is positive, so the direction stays: $m \leq 20 \div 4$, giving $m \leq 5$

(b) **Instructor-judged graph.** Filled circle at 5, shaded to the left.

4. Solve and graph $n + 9 > 2$

Solution: (a) Subtract 9 from both sides: $n > 2 - 9$, so

$$n > -7$$

(b) **Instructor-judged graph.** Open circle at -7 , shaded to the right. Both printed numbers are positive and the boundary is not — check the circle sits left of zero.

5. Solve and graph $\frac{p}{3} > 2$

Solution: (a) Multiply both sides by 3, which is positive, so the direction stays: $p > 2 \times 3$, giving $p > 6$

(b) **Instructor-judged graph.** Open circle at 6, shaded to the right.

6. Solve and graph $-2x \geq 10$

Solution: (a) Divide both sides by -2 . The divisor is negative, so the sign turns around: $x \leq 10 \div (-2)$, giving $x \leq -5$

(b) **Instructor-judged graph.** Filled circle at -5 , shaded to the left. Check: $x = -6$ gives $-2(-6) = 12 \geq 10$, and $x = 0$ gives 0, which is not.

7. Solve and graph $t - 6 \leq -1$

Solution: (a) Add 6 to both sides: $t \leq -1 + 6$, so

$$t \leq 5$$

(b) **Instructor-judged graph.** Filled circle at 5, shaded to the left. The right-hand side was negative and the boundary is positive.

8. Solve and graph $7w < -21$

Solution: (a) Divide both sides by 7. Seven is positive, so the direction stays even though the answer is negative: $w < -21 \div 7$, giving $w < -3$

(b) **Instructor-judged graph.** Open circle at -3 , shaded to the left. A negative answer is not by itself a reason to turn the sign around.

9. Solve and graph $-\frac{k}{4} < 2$

Solution: (a) Multiply both sides by -4 . The multiplier is negative, so the sign turns around: $k > 2 \times (-4)$, giving $k > -8$

(b) **Instructor-judged graph.** Open circle at -8 , shaded to the right. Check: $k = 0$ gives $0 < 2$, which works, and 0 is in the shaded part.

10. Solve and graph $15 \geq 3r$

Solution: (a) The letter starts on the right. Divide both sides by the positive 3: $5 \geq r$, which read from the letter outwards is $r \leq 5$

(b) **Instructor-judged graph.** Filled circle at 5, shaded to the left. Reading $5 \geq r$ backwards as $r \geq 5$ is the trap here.

11. Solve and graph $-5s < 35$

Solution: (a) Divide both sides by -5 ; the divisor is negative, so the sign turns around: $s > 35 \div (-5)$, giving $s > -7$

(b) **Instructor-judged graph.** Open circle at -7 , shaded to the right.

12. Solve and graph $-3v \geq 18$, then justify the turned-around sign.

Solution: (a) Divide both sides by -3 . The divisor is negative, so the sign turns around: $v \leq 18 \div (-3)$, giving $v \leq -6$

(b) **Instructor-judged graph.** Filled circle at -6 , shaded to the left.

(c) **Instructor-judged.** A correct response tests numbers rather than quoting the rule. Take $v = -10$ from the shaded part: $-3(-10) = 30$, and $30 \geq 18$, so -10 belongs in the solution. Take $v = 0$ from the other side: $-3(0) = 0$, and $0 \geq 18$ is false, so 0 does not belong. Multiplying by a negative number reverses the order of every pair of numbers, which is why the sign must turn around. Full credit needs at least one number tested in the original inequality and the conclusion drawn from it.
